

Jimmy He

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Work Authorization: U.S. Citizen

Master's CS student at Boston University seeking Summer 2026 Software Engineering internships. Experience building full-stack applications with React, Next.js, TypeScript, Python, Flask, PostgreSQL, REST APIs, and Docker.

Education

Boston University

M.S. in Computer Science; Dean's Scholarship Recipient

Boston, MA

Expected Dec 2026

University of Wisconsin-Madison

B.S. in Computer Science and Data Science; Certificate in Digital Studies

Madison, WI

May 2025

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Flask, pandas, NumPy, matplotlib, seaborn, Material UI

Databases & Tools: PostgreSQL, Prisma, Git, Docker, GCP, GitHub Actions, Linux, Jira

Testing & Practices: Jest, React Testing Library, REST APIs, RBAC, CI/CD, Agile/Scrum, Object-Oriented Design

Experience

Full Stack Developer | UW-Madison x NUU Mobile Industry Capstone

Jan 2025 – May 2025

React, Flask, PostgreSQL, Python, Docker, Jira, Git

Madison, WI

- Built a full-stack customer analytics dashboard for NUU Mobile using React, Flask REST APIs, PostgreSQL, and Docker to analyze customer behavior and churn trends
- Developed interactive dashboard features including data visualizations, filtering, aggregation, and client side pagination for exploring customer analytics data
- Integrated Flask API endpoints with PostgreSQL backed analytics queries to serve structured customer data to the React frontend
- Collaborated in a 4-person Agile team using Jira, Git, sprint planning, code reviews, and stakeholder demos throughout a 4-month development cycle
- Presented working software to NUU Mobile stakeholders and incorporated sprint feedback into backlog updates and dashboard improvements

Research Analyst | Customer Churn Prediction Study

May 2025 – Present

Python, pandas, NumPy, matplotlib, seaborn, Data Visualization

Remote

- Contributed to a research paper analyzing customer churn prediction using NUU Mobile device engagement data
- Created exploratory data visualizations using Python, pandas, matplotlib, and seaborn to analyze user behavior patterns, churn-related trends, and model performance outcomes
- Interpreted experimental results and translated visual findings into written research sections explaining key trends and retention related insights
- Reviewed manuscript content for clarity, consistency, and alignment between figures, analysis, and research conclusions
- Collaborated with research contributors to communicate data-driven findings for customer retention and deployment-oriented evaluation

Projects

Symporter Research Matching Platform | Client-Based Full-Stack Project

Jan 2026 – May 2026

Next.js, TypeScript, React, Prisma, PostgreSQL, Clerk, Jest, React Testing Library

- Built product features for a deployed research matching platform from client-provided Figma designs, implementing role aware navigation, dashboards, messaging flows, protected routes, and backend validation
- Developed responsive frontend components using Next.js, React, TypeScript, and Material UI for student and hiring-manager workflows
- Implemented authentication, route protection, and role based access control using Clerk and Next.js middleware to secure protected application pages
- Added server-side identity validation and ownership checks for Prisma/PostgreSQL API mutations to improve data reliability and access control

Legends: Monsters & Heroes | Java, OOP, Design Patterns

Nov 2025 – Jan 2026

- Designed a modular Java RPG game engine using Strategy, Factory, and Observer patterns across 40+ classes to support extensible game logic
- Implemented a finite state machine to manage combat, transactions, movement, and multi-entity interactions
- Built a data-driven configuration system with custom parsers to load game entities and settings without recompilation
- Separated game logic from rendering through an event-driven architecture, improving maintainability and testability